

Markscheme

Mathematics: analysis and approaches
Practice question bank

96. (a)

It can be proven by induction that $\sum_{r=1}^n r^2 = \frac{n(n+1)(2n+1)}{6}$ for every positive integer n .

Using this result, find the value of $\sum_{r=1}^{10} r^2$.

substitute $n = 10$: $\frac{10(11)(21)}{6}$ (M1)

$= 385$ A1

[5 marks]